

2023



Boosting Productivity, Optimizing Applications, Reducing Development Time Using AWS AML with Verisk

Learn how data analytics firm Verisk reduced commercial software licensing costs and optimized its applications while boosting productivity using AWS AML.

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Optimized
application performance

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Modernized
legacy commercial OS infrastructure

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development time

Improved

Overview

[Verisk Analytics Inc.](#) (Verisk), a leading strategic data analytics and technology partner to the global insurance industry, was seeking a way to improve its application architecture and reduce dependencies on commercial operating system (OS) licensing. Although the company had previously migrated its on-premises data centers to Amazon Web Services (AWS), it was still running migrated workloads in the cloud on commercial OSs as it had done in its data centers and was not taking advantage of “true AWS Cloud optimization.”

By using [AWS Application Modernization Lab](#) (AWS AML)—a three-phase AWS program focused on upskilling and accelerating customer cloud modernization strategies—the Verisk software, data, infrastructure, security, and DevOps engineering teams learned the best-practice cloud modernization skills. With these key learnings, Verisk has continued to apply best practices to remove the need for commercial licensing, modernize existing solutions, and develop new products.



Opportunity | Using AWS AML to Boost Productivity and Reduce Commercial OS Dependencies for Verisk

Verisk was founded in 1971 as a private adjunct company to insurance agencies. Since then, the company has grown into a multinational firm serving the global insurance industry with strategic data insights and risk analyses. Several years before the AWS AML project started, Verisk had taken steps toward cloud adoption, migrating from physical data centers to [Amazon Elastic Compute Cloud](#) (Amazon EC2), which provides secure, resizable compute capacity for virtually any workload; however, the cost and inflexibility associated with commercial OS licensing remained.

Despite having its data centers and applications on AWS, Verisk had not fully optimized its legacy commercial OS technology stack with all the appropriate AWS solutions to achieve ideal cost efficiency, scalability, and performance. So the company's Casualty Solutions team connected with the AWS AML team. "The AWS AML program was our attempt at using the AWS infrastructure properly," says Christopher Queenan, vice president of business technology at Verisk. "We wanted to learn how to increase our usage of Amazon Linux-based OSs and optimize our suite of solutions with guidance and upskilling from the AWS AML team."

The AWS AML hands-on lab portion of the engagement took about 8 weeks to complete after the AWS AML Technical Discovery phase. Two teams from Verisk worked on the project, one strictly on the development side and the other on the DevOps side. In the first phase, the company worked alongside AWS solutions architects to dive into the existing architecture. "The AWS AML team did a great job going through a deep dive of the entire application architecture," says Linto Mathew, assistant vice president of business technology for the Casualty Solutions team at Verisk. "We could immediately see how those initial deep-dive questions eventually led to delivering the ideal final product for us."

Solution | Using the AWS AML Playbook to Accelerate Modernization

After the lab completion, the team transitioned to the final phase: the AWS AML Phase 2 DIY Modernization. In this phase, the modernization of Verisk's Liability Navigator continued through production deployment. Alongside the AWS AML team, the Verisk development team chose the solutions that it believed worked well for modernizing the legacy application. Liability Navigator was modernized to run on a serverless infrastructure using [AWS Lambda](#), a serverless, event-driven compute service, thereby removing the need for and dependencies on commercial OSs.

Completing the migration from the SQL Server single-tenant environment to a multitenant application boosted performance tremendously. The multitenant application was powered by [Amazon DynamoDB](#), a fast, flexible NoSQL database service for single-digit millisecond performance at any scale. With the modernized application, Verisk saw improvement in its overall productivity by staying on a single instance while serving many customers. By going serverless, the company also saved resources and improved performance.

After gleaned technical learnings using AWS AML, Verisk applied these techniques to build products that match using AWS technologies. In product development, it continues to use AWS Lambda and [Amazon ElastiCache](#)



[Service](#) (Amazon ECS), which simplifies the deployment, management, and scaling of containerized applications. Several new products use architecture and infrastructure ideas that came out of the AWS AML project. One example is Discovery Navigator, a medical information retrieval tool, powered by natural language processing, that helps expedite insurance claim resolution. “Modernization best practices we learned through the training could be applied to real-world scenarios,” says Anilkumar Kukkal, development manager at Verisk. “We use that knowledge in our cloud center of excellence and all our current and downstream projects.”

During the AWS AML program, the Verisk DevOps team learned about the [AWS Cloud Development Kit](#) (AWS CDK), a software development framework that accelerates cloud development using familiar programming languages to model applications. Receiving sufficient training in AWS CDK, the company is now using it across many of its pipelines. Using AWS CDK in conjunction with Python, Verisk accelerated its serverless deployments. Another benefit was the time savings from the reusable frameworks that were developed during the AWS AML project. “Working alongside the AWS AML team, we built libraries that we can reuse for future application development,” says Ranjith Panakal, a technical architect at Verisk. “This has saved us a lot of time.”

After completing the AWS AML project, Verisk continued upskilling its teams and developing best practices by promoting AWS education among its workforce. Everyone who worked on the project became AWS Certified through the AWS AML program. Now, it has become an ongoing initiative for the company to add AWS Certifications each year. This initiative has boosted employee morale and engagement as the company works toward completing these training events. “AWS AML shaped our goals of attaining one new AWS Certification per year per employee. Getting AWS Certified means gaining a new mindset about new products and new solutions,” says Mathew. “We make sure these learnings benefit our individual employees and in turn the whole company.”

Outcome | Continuing to Transform and Modernize Architecture

Experiencing the benefits of successfully modernizing its legacy commercial OS applications using AWS AML, Verisk plans to continue building and transforming its architecture with future projects. The company believes strongly that the learnings gained through AWS AML will encourage more growth and cost savings down the line.

“Using AWS AML, we can empower our employees,” says Queenan. “They gain skills and emerge with a deliverable that benefits the company. Our company wins and our employees win using AWS AML.”

About Verisk Analytics Inc.

Verisk, a leading strategic data analytics and technology partner to the global insurance industry, empowers clients to strengthen efficiency, improve underwriting/claims outcomes, combat fraud, and make informed decisions about global risks.

AWS Services Used

Modernizing your business applications is an inevitable aspect of doing business. With AWS, you get to decide on how to migrate your applications and at what pace, leveraging the industry's most reliable infrastructure with the deepest set of services.

[Learn more »](#)

AWS CDK

AWS Cloud Development Kit (AWS CDK) accelerates cloud development using common programming languages to model your applications.

[Learn more »](#)

Amazon EC2

Amazon Elastic Compute Cloud (Amazon EC2) offers the broadest and deepest compute platform, with over 750 instances and choice of the latest processor, storage, networking, operating system, and purchase model to help you best match the needs of your workload.

[Learn more »](#)

AWS Lambda

AWS Lambda is a compute service that runs your code in response to events and automatically manages the compute resources, making it the fastest way to turn an idea into a modern, production, serverless applications.

[Learn more »](#)

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